Application of Botulinum Toxin at the Yonsei Point for the Treatment of Gummy Smile: A Randomized Controlled Trial

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Background: Demand for less-invasive procedures for treating gummy smile, such as botulinum toxin A injections, has increased substantially over the years. Meanwhile, the optimal injection site for botulinum toxin A injection is debated. The authors aimed to investigate the efficacy of botulinum toxin A injection at the Yonsei point for treating gummy smile.

Methods: In this double-blind, single-site, randomized clinical trial, healthy participants with a gummy smile (anterior gingival exposure of ≥3.0 mm) were enrolled and randomized (1:1 ratio) into two groups. The experimental group was administered 6 U of botulinum toxin A at the Yonsei point (a single-site injection of 3 U to the right Yonsei point and 3 U to the left Yonsei point), and the control group received the same dose in the bilateral levator labii superioris alaeque nasi muscle sites. The patients were assessed at baseline and 4, 12, 24, and 48 weeks after the first injection using a digital vernier caliper.

Results: A total of 49 participants were enrolled. Anterior and bilateral posterior gingival exposure were reduced at 4, 12, and 24 weeks ($P \leq 0.05$) and returned to baseline at 48 weeks in both groups; there was no difference between the groups at these time points. The increase in satisfaction among patients was significant, and few adverse events were observed.

Conclusion: Both the Yonsei point and the levator labii superioris alaeque nasi muscle site can be used as botulinum toxin A injection sites for treating gummy smile. *(Plast. Reconstr. Surg.* 153: 711e, 2024.)

CLINICAL QUESTION/LEVEL OF EVIDENCE: Therapeutic, I.

A gummy smile, also known as a gingival smile or an excessive gingival display, is characterized by the exposure of more than 3 mm of the gingiva upon smiling.[1] The prevalence of gummy smile varies from 2.0% to 10.6% in people from different races and ethnicities.[2] An excessive gingival display may interfere with self-esteem and psychologic status; thus, these individuals may seek aesthetic intervention. Furthermore, most orthodontists and dentists consider a gingival smile a significant risk factor for dental treatment.[3] Gummy smile can be corrected by gingivoplasty, bone resection, orthognathic surgery, or lip repositioning; however, the surgical trauma, cost, and painful postoperative period often deter individuals from pursuing these treatments.[1]

Muscular hyperactivity of the upper lips is a key anatomic determinant in the genesis of the

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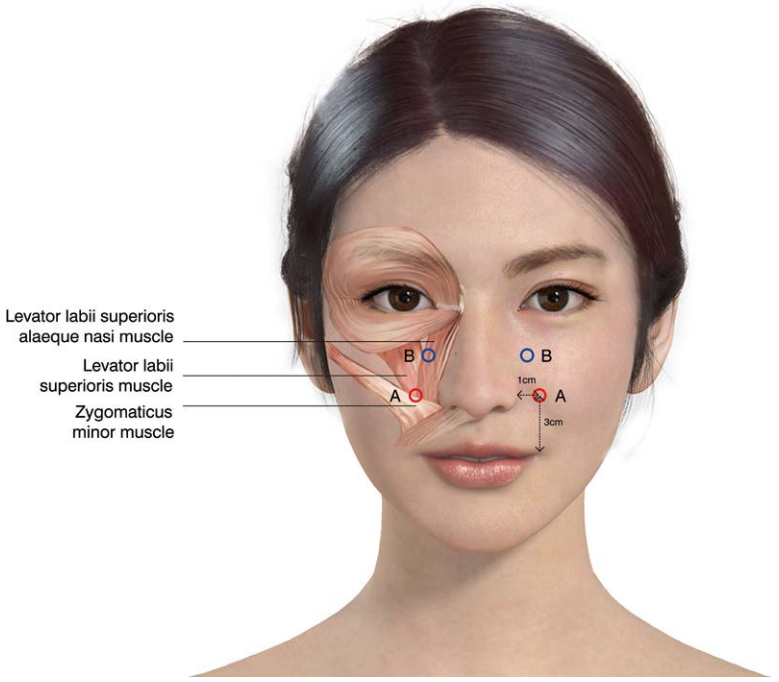


Fig. 1. Illustration of specific muscle injection points. *Point A* is the injection site at the Yonsei point, localized 1 cm lateral to the alae nasi horizontally and 3 cm above the lip line vertically, which targets the whole of the levator labii superioris alaeque nasi, levator labii superioris, and zygomaticus minor muscles. *Point B* is the injection site at the bilateral levator labii superioris alaeque nasi, located at the uppermost part of the nasolabial fold.

gummy smile. According to Peck et al.,[4] people with gummy smile had nearly 20% greater facial muscular capacity to raise the upper lip when smiling than those with average lip lines. The smile is mainly dictated by muscles' points of origin and insertion, as well as the direction of the force vectors dictated by their contraction and eventual relaxation. Levator labii superioris alaeque nasi (LLSAN), levator labii superioris (LLS), zygomaticus minor (ZMi), and zygomaticus major muscles are the major muscles that control lifting and elevation of the upper lip during smiling.[1,5]

Botulinum toxin A can reversibly prevent the release of the neurotransmitter acetylcholine from axon endings at the neuromuscular junction and cause temporary flaccid paralysis. Since 1970, botulinum toxin A therapy has been investigated for the treatment of various conditions associated with muscle contraction, pain, or cosmetic use owing to its clinical safety and efficacy.[6] Botulinum toxin A was first applied for gummy smile by Polo[6] in 2005, and it had a tangible effect by reducing the contractility of the muscles that raise and elevate the upper lip during smiling. Since then, botulinum toxin A has been regarded as the first-choice technique for gummy smile treatment.[1,6–14]

Treatment efficacy of botulinum toxin A for gummy smile varies from study to study, with gingival exposure (GE) improvement rate ranging from 62.06% to 98%.[1,6,12,15] Studies have suggested that injection sites or target muscles may affect the treatment effects and the smile aesthetics.[16] Most studies recommend injecting botulinum toxin A into the LLSAN muscle.[8,17,18] In addition, the Yonsei point (Fig. 1, point A), proposed by Hwang et al.,[5] has been recommended as a safe and repeatable injection point.[19–21] The Yonsei point, located 1 cm lateral to the alae nasi horizontally and 3 cm above the lip line vertically, is the center of this triangle; thus, injection in this point targets the LLSAN, LLS, and ZMi.[5] However, the adaptation of the Yonsei point as an injection point to treat gummy smile has been rarely explored in the literature.

We compared the efficacy of botulinum toxin A injection at the Yonsei point with injection

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at the LLSAN muscle site for treating gummy smile. Patient satisfaction and side effects were analyzed.

PATIENTS AND METHODS

Study Design and Participants

This double-blind, single-site, parallel randomized clinical trial was conducted at the Second Clinical Division of Peking University School and Hospital of Stomatology in Beijing, China, between October of 2020 and October of 2021. This trial followed the Consolidated Standards of Reporting Trials (CONSORT) statement[22] and is registered at clinicaltrials.gov (identification no. NCT04527744). The ethics committee of the institutional review board of Peking University Health Science Center approved this study (approval no. PKUSSIRB-202056097) and written informed consent was obtained from all patients.

Public online and offline recruitment materials (**See Document, Supplemental Digital Content 1**, which shows the translated version of the recruitment materials, ***http://links.lww.com/PRS/G554***) were used to enroll participants in this study. Individuals who had a gummy smile or previously sought treatment but did not achieve expected results were recruited. The inclusion criteria were age between 18 and 60 years and anterior GE 3.0 mm or greater upon unrestricted full smiling. Exclusion criteria were contraindications to botulinum toxin A (eg, keloidal scarring, neuromuscular disorders, botulinum toxin allergies, body dysmorphic disorder, pregnancy, breastfeeding, amyotrophic lateral sclerosis, myopathies); previous disease or treatment affecting the position of the upper lip or gingiva; a history of botulinum toxin A injection to the head or neck region within 1 year; received or receiving active orthodontic treatment, including vertical dimension treatment, such as for extrusion or intrusion; or periodontal disease.

The sample size was estimated by comparing anterior GE using the PASS 11.0 software package (NCSS) with a power of 90% and an α error of 5%. The estimated sample size was 44, with anterior GE 1.15 ± 0.74 mm in the experimental group, according to the literature,[21] and 2.84 ± 2.31 mm in the control group, according to our preliminary experiment. Considering loss to follow-up, the estimated sample size was expanded by 10%. Thus, the final sample size was at least 48 cases (matched 1:1; 24 in each group).

Interventions

A random sequence was generated by Z.H.T. using a computerized random number generator (R version 4.1.2). Participants were enrolled (X.G.) and randomly allocated into two groups (experimental and control groups, using a 1:1 ratio). Neither the participants nor the experimenter who assessed the outcomes was blinded to the grouping. The same botulinum toxin A (Botox; Allergan) and a uniform botulinum toxin A injection technique[23] were used for all participants. One hundred units of botulinum toxin was diluted with 2.5 mL of 0.9% sterile saline solution, and a 27-gauge insulin syringe was used for injection. No anesthesia was given during the procedure. All injections were performed by the same operator (X.G.).

The patients in the experimental group received 6 U of botulinum toxin: a single-site injection of 3 U was given to the right Yonsei point,[5] and 3 U to the left Yonsei point, which is localized 1 cm lateral to the ala nasi horizontally and 3 cm above the lip line, vertically (Fig. 1, point A).

The patients in the control group were administered the same dose (ie, 3 U at the right LLSAN muscle site and 3 U at the left LLSAN muscle site). The injection points were located at the muscle bulge at the uppermost part of the nasolabial fold (Fig. 1, point B).

Outcome Measurements

Standardized measurements were performed at baseline and during follow-up (ie, 4, 12, 24,

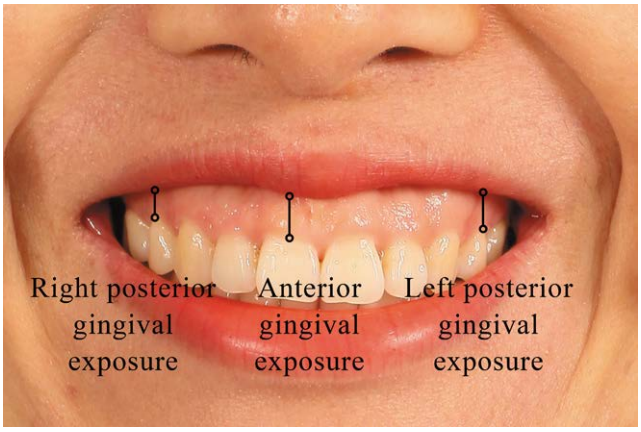


Fig. 2. Reference points for primary and main secondary outcome measures in a 33-year-old woman. Anterior gingival exposure was based on the distance between the superior margin of the right incisor and the lower margin of the upper lip. Right and left posterior gingival exposure was defined as the distance between the superior margin of the bilateral first premolars and the lower margin of the upper lip.

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Fig. 3. Illustrative images of immediate results and follow-up sessions of a 55-year-old woman with injection site at the Yonsei point. (*Above*) Before injection. (*Center*, *left*) Four weeks after injection. (*Center*, *right*) Twelve weeks after injection. (*Below*, *left*) Twenty-four weeks after injection. (*Below*, *right*) Forty-eight weeks after botulinum toxin A injection.

and 48 weeks postinjection). All measurements were performed by Z.W. using a digital vernier caliper. To increase the comparability and repeatability of the results throughout the study period, full spontaneous smiling was used for GE measurement.[2,17,18,24] Participants were exposed to funny jokes, statements, singing, or dancing to induce full, unrestricted, spontaneous smiling, as described by Sarver and Ackerman.[3] The measurements were taken individually during three different spontaneous smiles to ensure maximal smile for each measured area.

The primary outcome measure was posttreatment anterior GE (in millimeters) at 4 weeks, defined as the distance between the lower margin of the upper lip and the superior margin of the right incisor (Fig. 2). The secondary outcome measures included the anterior GE at 12, 24, and 48 weeks postinjection and the bilateral posterior GE (the distance from the bilateral first premolars and the lower margin of the upper lip during maximal smiling, respectively) at 4, 12, 24, and 48 weeks postinjection. The severity and type of gummy smile[1] were also evaluated by measuring anterior and bilateral posterior GE.

The etiologies were also investigated. Upper-lip mobility was assessed across the maxillary right central incisor from rest to maximal smiling to detect the presence of upper-lip hypermobility. When upper-lip mobility during smiling was greater than 8 mm, hypermobility of the upper lip was noted.[25–27] A short clinical crown was defined

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as a width/length ratio of maxillary incisor teeth of greater than 0.85 to assess alterations in anatomic features.[28,29] Anterior dentoalveolar extrusion was assessed by overbite and overjet of the anterior teeth. Deep overbite was diagnosed when the vertical overlap between the upper and lower incisors was greater than one-third of the lower incisor tooth height; deep overjet was considered when the extent of horizontal (anteroposterior) overlap of the maxillary central incisors over the mandibular central incisors was 3 mm or greater. Short upper lips were characterized when the upper lip was less than 15 mm, measured from the subnasale to the lower border of the upper lip.[28] Tooth display at rest was also measured and was defined as the distance between the inferior margin of the right incisor and the lower margin of the upper lip when the participant was at rest. Patients with short upper lips or maxillary hyperplasia present with excessive tooth exposure at repose.[30] Tooth display and the length of the upper lip and right medial incisor were also measured during follow-up to assess the static alteration.

An anonymous questionnaire was used (A.-R.Z.) to assess patient satisfaction, side effects, willingness to repeat the therapy, and perceived improvement in the nasolabial fold. (**See Document, Supplemental Digital Content 2**, which shows the questionnaire measuring satisfaction, side effects,



Fig. 4. Illustrative images of immediate results and follow-up sessions of a 24-year-old woman with injection site at the bilateral levator labii superioris alaeque nasi. (*Above*) Before injection. (*Center*, *left*) Four weeks after injection. (*Center*, *right*) Twelve weeks after injection. (*Below*, *left*) Twenty-four weeks after injection. (*Below*, *right*) Forty-eight weeks after botulinum toxin A injection.

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Table 1. Baseline Characteristics According to Treatment Group

Characteristics	Experimental Group (n = 24)	Control Group (n = 25)	P
Female[a]	23 (95.8)	19 (76.0)	0.098
Age, yrs[b]	30.0 ± 9.4	30.3 ± 8.3	0.822
Severity of gummy smile[c]			0.587
3 mm ≤ anterior GE < 5 mm	2 (8.3)	1 (4)	
5 mm ≤ anterior GE < 7 mm	11 (45.8)	15 (60.0)	
Anterior GE ≥7 mm	11 (45.8)	9 (36.0)	
Type of gummy smile[c]			0.161
Anterior	1 (4.2)	3 (12.0)	
Mixed	23 (95.8)	19 (76.0)	
Asymmetric	0	3 (12.0)	
Hypermobility of the upper lip, mm[b]	11.3 ± 1.6	11.4 ± 1.8	0.745
Width/length ratio of maxillary incisor teeth[b]	0.9 ± 0.1	0.9 ± 0.1	0.507
Overbite[a]	18 (75.0)	18 (72.0)	0.812
Overjet[a]	16 (66.7)	19 (76.0)	0.538
Length of the upper lip, mm[b]	22.5 ± 2.1	23.6 ± 2.4	0.320
Tooth display at rest, mm[b]	4.8 ± 1.9	4.2 ± 2.3	0.704
Pain at the procedure[d]	2 (2-2)	2 (2-2)	0.564

GE, gingival exposure.
[a]Indicator values are presented as the number of participants (%) based on the Pearson chi-square test.
[b]Indicator values are presented as mean ± SD based on an independent-samples t test.
[c]Indicator values are presented as the number of participants (%) based on the Fisher exact test.
[d]Indicator values are presented as median (25th percentile, 75th percentile) based on the Mann-Whitney U test. Nonparametric analysis.

and willingness to repeat the therapy after the procedure, ***http://links.lww.com/PRS/G555***.) In addition, facial images and videos were captured for documentation. Photographs of patients before and after botulinum toxin injection are presented in Figures 3 and 4.

Statistical Analysis

The WJX process (Ranxing Information Technology) was used to examine data entry. The Shapiro-Wilk test was used to determine whether the data conformed to normal distribution. Mean ± SD values were used to describe normally distributed numerical values. Paired t test was used to compare preoperative with postoperative effects over time points. Independent-samples t test was used to compare normally distributed variables in two independent groups. Cohen d was used to indicate the standardized difference between the two means. Mann-Whitney U test was used for nonnormally distributed numerical values. Chi-square test was used to compare categorical differences. Fisher exact test was used when the cell numbers were less than five. SPSS software version 20.0 (IBM) was used for all statistical analyses. A statistically significant difference was defined as $P \leq 0.05$.

RESULTS

A total of 49 patients, including seven men and 42 women (mean age, 30.13 years; range, 20.1 to 60.0 years), were included and randomly assigned to the experimental group (n = 24) or the control group (n = 25). The distribution of baseline characteristics in groups was consistent (Table 1; $P > 0.05$). Forty-four patients participated in the 4-week follow-up, 37 in the 12-week follow-up, 37 in the 24-week follow-up, and 27 in the 48-week follow-up. The CONSORT diagram of the study is shown in Figure 5. Some patients were lost to follow-up, mainly because of the COVID-19 pandemic. However, none of the patients withdrew because of treatment dissatisfaction.

The anterior GE was 6.8 ± 1.4 mm at baseline (49 participants), decreased to 4.2 ± 1.5 mm (44 participants; $P \leq 0.05$) at week 4, and then gradually increased to 5.1 ± 1.6 mm at week 12 (37 participants; $P \leq 0.05$), 6.0 ± 1.3 mm at week 24 (37 participants; $P \leq 0.05$), and 6.8 ± 1.2 mm at week 48 (27 participants; $P > 0.05$). These changes were seen in both groups at 4, 12, 24, and 48 weeks ($P > 0.05$; Table 2).

For the total sample and both groups separately, posttreatment bilateral posterior GEs were significantly reduced compared with pretreatment at weeks 4 and 12 ($P \leq 0.05$), and then returned to baseline levels by week 48 ($P > 0.05$; Table 2). An evaluation of bilateral posterior GEs revealed no significant difference between the two groups at any time point ($P > 0.05$; Table 2).

Tooth display at rest and upper lip length in both groups changed significantly at 4 and 12

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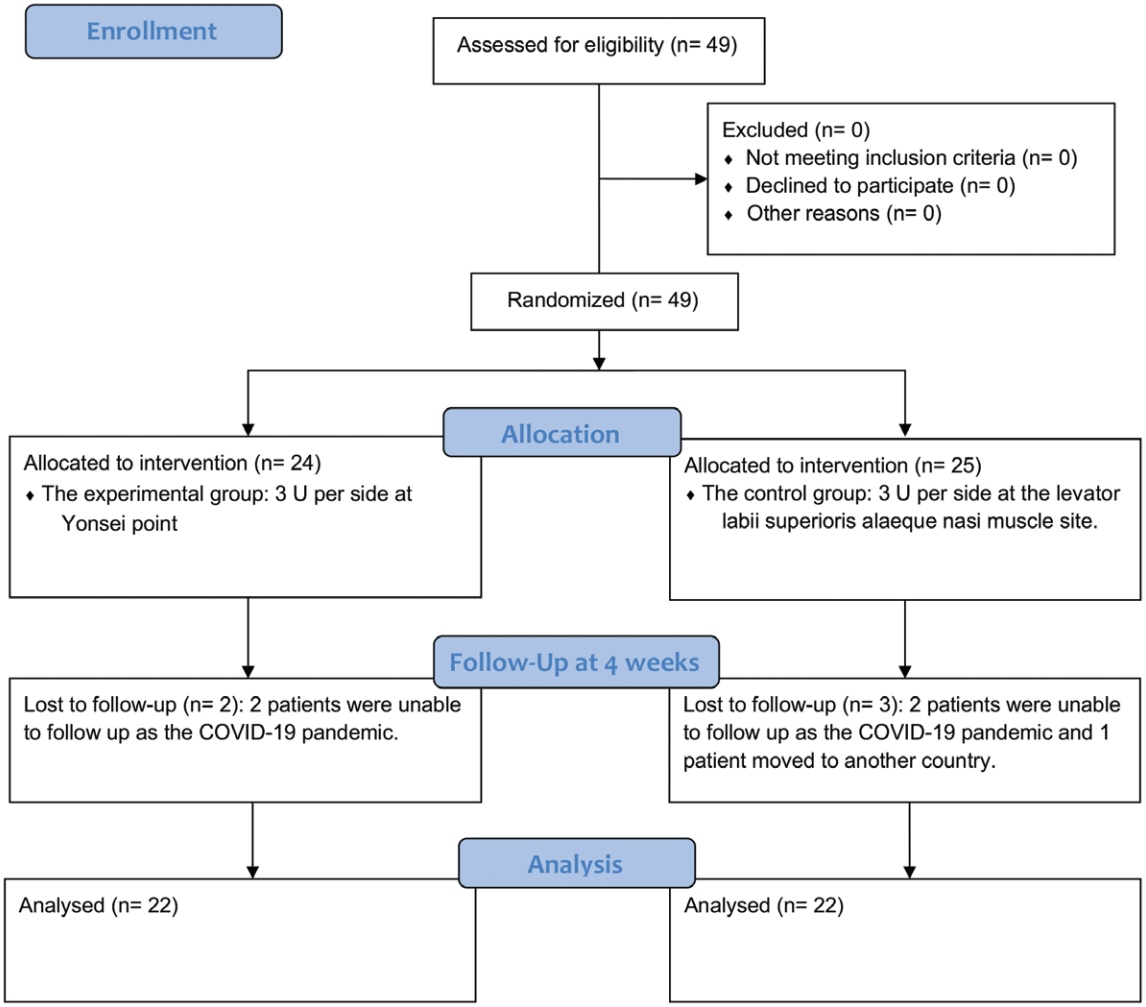


Fig. 5. CONSORT flow diagram.

weeks compared with baseline ($P \leq 0.05$) and returned to baseline at 24 weeks and 48 weeks; there were no differences between the two groups ($P > 0.052$ Table 2).

Subjective evaluations of participant satisfaction and side effects are shown in Table 3. [**See Video (online)**, which shows a follow-up interview with a patient after treatment.] At 4 weeks, 88.6% (39 of 44) of the patients thought their gummy smiles were improved. Among 39 patients who expressed a positive attitude toward the treatment effect, 30 were completely satisfied, and nine expected greater improvement. In addition, 18 out of 22 patients (88.6%) in the experimental group and 21 out of 22 patients (95.5%) in the control group thought their gummy smiles were improved after injection, without significant differences between the two groups ($P > 0.05$). Five patients believed the treatment effect could not be evaluated. No one judged the treatment as ineffective. The subjective improvement rate was 64.9% (24 of 37) at 12 weeks, and 20 individuals reported maintenance of the effect at 24 weeks. However, no participants reported effect maintenance at 48 weeks. No significant differences in the subjective improvement rates were found between the two groups ($P > 0.05$). Apart from improvement in gummy smile, 47.7% of participants (21 of 44) reported improvement in smoothness in the nasolabial fold. In this study, 20.45% of patients (nine of 44) had slight lip ptosis, 18.2% (eight of 44) had slight asymmetry, and 11.4% (five of 44) had slightly excessive upper lip, without significant differences between the two groups ($P > 0.05$; Table 3). All adverse effects were mild, could not be verified during clinical examination, and disappeared within 4 weeks without further intervention.

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Table 2. Follow-Up Measurements[a]

Measurements/Group	Baseline	4 Weeks	Δ	*P*[b]	12 Weeks	Δ	*P*[b]	24 Weeks	Δ	*P*[b]	48 Weeks	Δ	*P*[b]
Anterior GE													
Experimental	6.8 ± 1.4; (*n* = 24)	4.6 ± 1.5; (*n* = 22)	2.4 ± 1.1	0.000	5.4 ± 1.6; (*n* = 20)	1.6 ± 1.0	0.000	6.3 ± 1.4; (*n* = 18)	0.8 ± 1.2	0.011	7.0 ± 1.4; (*n* = 14)	0.0 ± 0.8	0.870
Control	6.7 ± 1.4; (*n* = 25)	3.7 ± 1.4; (*n* = 22)	2.9 ± 1.2	0.000	4.9 ± 1.5; (*n* = 17)	1.6 ± 1.2	0.000	5.8 ± 1.2; (*n* = 19)	1.0 ± 0.7	0.000	6.5 ± 1.0; (*n* = 13)	0.0 ± 0.6	0.996
P[c]	0.728	0.798	0.597	—	0.290	0.691	—	0.497	0.312	—	0.386	0.333	—
Cohen *d*	—	0.620	−0.434	—	0.321	0.000	—	0.384	−0.205	—	0.408	0.000	—
Right posterior GE													
Experimental	6.0 ± 1.6	4.0 ± 1.7	2.0 ± 1.1	0.000	5.0 ± 1.9	1.1 ± 1.4	0.003	5.8 ± 1.7	0.3 ± 1.1	0.210	5.9 ± 2.1	0.2 ± 1.2	0.574
Control	4.6 ± 2.1	3.4 ± 2.2	1.4 ± 1.1	0.000	4.3 ± 1.5	1.0 ± 1.5	0.001	4.2 ± 1.9	0.5 ± 1.4	0.112	5.6 ± 2.1	−0.3 ± 1.2	0.691
P[c]	0.146	0.313	0.667	—	0.566	0.951	—	0.387	0.159	—	0.891	0.970	—
Cohen *d*	—	0.305	0.545	—	0.405	0.069	—	0.886	−0.158	—	0.143	0.417	—
Left posterior GE													
Experimental	5.7 ± 1.6	4.2 ± 1.9	1.6 ± 1.1	0.000	4.8 ± 1.8	1.0 ± 1.2	0.013	5.4 ± 1.7	0.5 ± 0.7	0.144	5.7 ± 2.0	0.1 ± 1.2	0.426
Control	4.5 ± 1.9	2.9 ± 1.8	1.6 ± 0.9	0.000	3.9 ± 1.4	1.1 ± 1.3	0.003	4.1 ± 1.5	0.4 ± 0.8	0.061	5.3 ± 1.5	−0.2 ± 1.0	0.581
P[c]	0.228	0.972	0.315	—	0.398	0.862	—	0.715	0.736	—	0.311	0.635	—
Cohen *d*	—	0.702	0.000	—	0.552	−0.080	—	0.812	0.133	—	0.225	0.271	—
Tooth display at rest													
Experimental	4.8 ± 1.9	4.6 ± 1.8	0.4 ± 0.7	0.007	4.6 ± 1.6	0.6 ± 1.0	0.021	4.7 ± 1.9	0.4 ± 1.0	0.092	4.9 ± 1.8	0.2 ± 0.4	0.064
Control	4.2 ± 2.3	3.4 ± 1.8	0.4 ± 0.9	0.044	3.1 ± 1.7	0.3 ± 0.8	0.039	3.9 ± 1.5	−0.2 ± 1.3	0.534	3.1 ± 2.1	1.1 ± 0.8	0.567
P[c]	0.704	0.860	0.218	—	0.850	0.134	—	0.360	0.515	—	0.745	0.145	—
Cohen *d*	—	0.666	−0.124	—	0.911	0.328	—	0.469	0.515	—	0.923	−1.440	—
Upper lip length at rest													
Experimental	22.5 ± 2.1	22.9 ± 1.7	−0.8 ± 0.5	0.000	22.8 ± 1.8	−0.6 ± 0.4	0.000	22.5 ± 1.7	−0.3 ± 0.7	0.133	22.2 ± 1.8	−0.1 ± 0.3	0.285
Control	23.6 ± 2.4	24.0 ± 2.6	−0.4 ± 0.6	0.013	24.0 ± 2.8	−0.4 ± 0.4	0.001	24.3 ± 2.3	−0.2 ± 0.8	0.285	23.8 ± 2.6	0.1 ± 0.4	0.581
P[c]	0.320	0.71	0.445	–	0.059	0.612	–	0.523	0.863	–	0.107	0.573	–
Cohen *d*	—	−0.501	−0.724	—	−0.519	−0.500	—	−0.886	−0.133	—	−0.721	−0.569	—
Medial incisor length													
Experimental	9.3 ± 1.0	9.4 ± 1.0	−0.1 ± 0.2	0.149	9.4 ± 1.1	−0.1 ± 0.5	0.306	9.3 ± 1.2	−0.1 ± 0.2	0.113	9.1 ± 1.0	0.0 ± 0.4	0.918
Control	8.9 ± 0.9	9.0 ± 1.1	0.0 ± 0.3	0.591	9.1 ± 1.0	−0.2 ± 0.6	0.205	8.9 ± 1.0	0.0 ± 0.2	0.518	9.0 ± 1.0	−0.1 ± 0.4	0.471
P[c]	0.588	0.203	0.140	—	0.934	0.607	—	0.802	0.754	—	0.752	0.940	—
Cohen *d*	—	0.381	−0.392	—	0.284	0.182	—	0.363	−0.500	—	0.100	0.250	—

GE, gingival exposure.

[a]Data are presented as mean ± SD.

[b]A paired-samples *t* test was used to compare the preoperative and postoperative effects. The mean value of these differences (ie, Δ) was obtained by subtracting the preoperative data from the postoperative data, and data of the patients who were lost to follow-up were excluded from the calculation.

[c]An independent-samples *t* test was used to compare the effects of both methods. Cohen *d* was used to indicate the standardized difference between the two means.

Table 3. Evaluation of Subjective Satisfaction and Side Effects[a]

Satisfaction and Side Effects	4 Weeks			12 Weeks			24 Weeks			48 Weeks		
	Experimental Group ($n = 22$)	Control Group ($n = 22$)	*P*	Experimental Group ($n = 20$)	Control Group ($n = 17$)	*P*	Experimental Group ($n = 18$)	Control Group ($n = 19$)	*P*	Experimental Group ($n = 14$)	Control Group ($n = 13$)	*P*
Satisfied with the treatment effects			0.262			0.232			0.787			1.000
Improvement												
Completely satisfied	12	18		11	13		9	11		1	0	
Pleased but expecting better improvement	6	3										
No improvement	0	0		4	1		8	5		9	8	
Cannot be evaluated	4	1		5	3		1	3		4	5	
Willingness to undergo a repeat procedure												
Willing	22	21	0.512	—	—	—	—	—	—	—	—	—
Maybe	0	1		—	—		—	—		—	—	
Unwilling	0	0		—	—		—	—		—	—	
Willingness to recommend this treatment to others												
Willing	21	19	0.657	—	—	—	—	—	—	—	—	—
Maybe	1	3		—	—		—	—		—	—	
Unwilling	0	0		—	—		—	—		—	—	
Smoothness of the nasolabial fold												
Good	3	1	0.654	—	—	—	—	—	—	—	—	—
Fair	9	8		—	—		—	—		—	—	
No improvement	6	5		—	—		—	—		—	—	
Not paying attention	4	8		—	—		—	—		—	—	
Improvement of the smile arc												
Good	7	5	0.440	—	—	—	—	—	—	—	—	—
Fair	4	8		—	—		—	—		—	—	
Slight lip ptosis	5	4	1.000	0	0	1.000	0	0	1.000	0	0	1.000
Slight asymmetry	5	3	0.742	0	0	1.000	0	0	1.000	0	0	1.000
Slight excessive upper lip	2	3	1.000	0	0	1.000	0	0	1.000	0	0	1.000

[a]Indicator values are presented as the number of participants. *P*: Fisher exact test was used to compare the effects of the two methods. —, indicator was not evaluated.

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DISCUSSION

This study compared the therapeutic effect of botulinum toxin A injection at the Yonsei point with the LLSAN muscle site in patients with gummy smiles. In previous studies, the LLSAN muscle was the main injection point when treating gummy smile.[1,8,12,17,18] The LLSAN muscle originates from the frontal process of the maxilla and extends downward into two sections—the internal part expands the nostrils; the external part is further down and joins with the LLS.[31] Sucupira and Abramovitz[12] administered botulinum toxin A in the LLSAN muscle of 52 patients and reported a satisfaction level of 9.75 on a 10-point scale. They also found that this injection site is associated with a minimum risk of complications compared with other injection sites. However, Mazzuco and Hexsel[1] indicated that injecting botulinum toxin A in the LLSAN muscle may not be sufficient. They claimed that the ZMi is more functionally involved than the other muscles of the upper-lip lift complex in the case of posterior gummy smile, and the conventional application of botulinum toxin A to relax the LLSAN muscle would cause unnecessary lowering of the lip in the medial region, leading to undesirable aesthetic and functional results. The ZMi originates from the zygomatic bone and is inserted into the upper lip (Fig. 1). The LLS is partially or entirely covered by the LLSAN and the ZMi insertions. All three muscles pass near a triangular region and converge at the area lateral to the ala of the nose. The Yonsei point, which is located 1 cm lateral to the alae nasi horizontally and 3 cm above the lip line vertically, is the center of this triangle.[5]

We hypothesized that injection of botulinum toxin A at the Yonsei point could provide improved treatment effects for gummy smile. However, we found no difference in GE reduction (anterior or posterior), side effects, or patient satisfaction when the botulinum toxin was injected at the Yonsei points or LLSAN muscle site. We speculated this could be attributed to botulinum toxin diffusion in the soft tissue.

In this study, the effect of botulinum toxin A injection lasted for more than 24 weeks, which is higher compared with data obtained by Chagas et al.[32] (the effect was stable for at least 8 weeks) and Zengiski et al.[33] (the effect was stable for at least 12 weeks, returning close to initial values within 24 weeks after application), but consistent with Polo,[17] who reported that GE gradually increased from 2 weeks after the injection and lasted for 24 weeks, after which GE gradually returned to its baseline values. This 8- to 12- to 24-week duration may also be achieved when using botulinum toxin A injections for other indications (eg, strabismus, chronic headache, facial wrinkles). Botulinum toxin A injection is widely used because it is microinvasive; has low risk with minimal side effects; and can be considered a low-cost, easy, and fast outpatient procedure that requires no downtime and has high efficacy rates.[34]

This study has a few limitations. This was a single-center study with a small sample size. In addition, the therapeutic dose used was relatively small (3 U on each side). There is substantial debate about the optimal dose of botulinum toxin A for the treatment of gummy smile.[1,11,12,18,35,36] According to our recent findings, 3 U per side is suitable for a gummy smile with the anterior GE less than 5 mm, but is not sufficient for GE of 5 mm or greater.[37] Thus, further research will be designed to explore injection doses for more appropriate treatment for gummy smile.

CONCLUSIONS

Botulinum toxin injection can reduce GE in patients with gummy smiles, achieving results that last for more than 24 weeks. However, we found no difference in GE reduction (anterior or posterior), side effects, or patient satisfaction between different injection sites (Yonsei point versus LLSAN muscle site).

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DISCLOSURE

The authors have no financial interest to declare in relation to the content of this article.

PATIENT CONSENT

Patients provided written informed consent for the use of their images.

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